

### **3. TRIPLE DES (3DES)**

#### **AIM**

To implement the Triple DES (3DES) algorithm using Python.

#### **Algorithm**

1. Start.
2. Import DES3 library.
3. Enter plaintext.
4. Enter a 24-character secret key.
5. Encrypt the plaintext using Triple DES.
6. Display the ciphertext.
7. Decrypt the ciphertext.
8. Display the original plaintext.
9. Stop.

#### **Python Code**

```
from Crypto.Cipher import DES3
from Crypto.Util.Padding import pad, unpad
import base64

plaintext = input("Enter the Plain Text: ")
key = input("Enter a 24-character Key: ").encode()
cipher = DES3.new(key, DES3.MODE_ECB)

encrypted = cipher.encrypt(pad(plaintext.encode(), DES3.block_size))
print("\nEncrypted Text :", base64.b64encode(encrypted).decode())

decrypted = unpad(cipher.decrypt(encrypted), DES3.block_size)
print("Decrypted Text :", decrypted.decode())
```

## Output

```
Enter the Plain Text: hello world
Enter a 24-character Key: 123456789012345678901234

Encrypted Text : SdHQCpbVRzk4JSGbnhUMLg==
Decrypted Text : hello world
```

## The result:

The above code is executed and the output is obtained successfully.